Introduction to ML and data science in chemistry

CAMLC Workshop 2026, Zaragoza

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Learning outcomes for this session

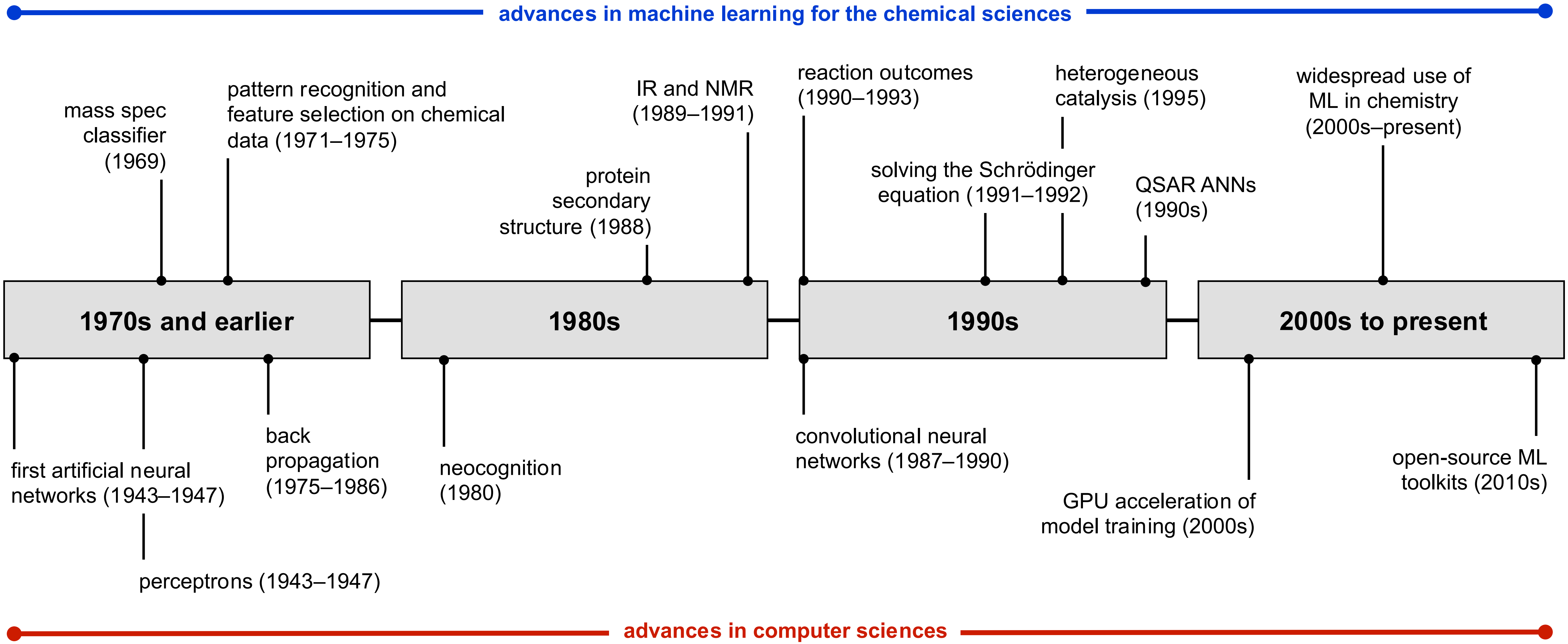
- By the end of this session, you should be able to:

- [ ] **Explain** the role and key applications of machine learning in chemistry.
- [ ] **Assess** the quality and limitations of chemical datasets, including bias and FAIR principles.
- [ ] **Interpret** chemical data types and distributions, and their impact on model selection and performance.
- [ ] **Distinguish** between regression and classification approaches for chemical problems.
- [ ] **Apply** core data preprocessing (*e.g.*, scaling, feature selection, dataset splitting) steps in a simple ML workflow.
- [ ] **Construct and evaluate** basic linear regression and classification models using chemical data.
- [ ] **Critically evaluate** the reliability and applicability of machine learning models in chemical research.

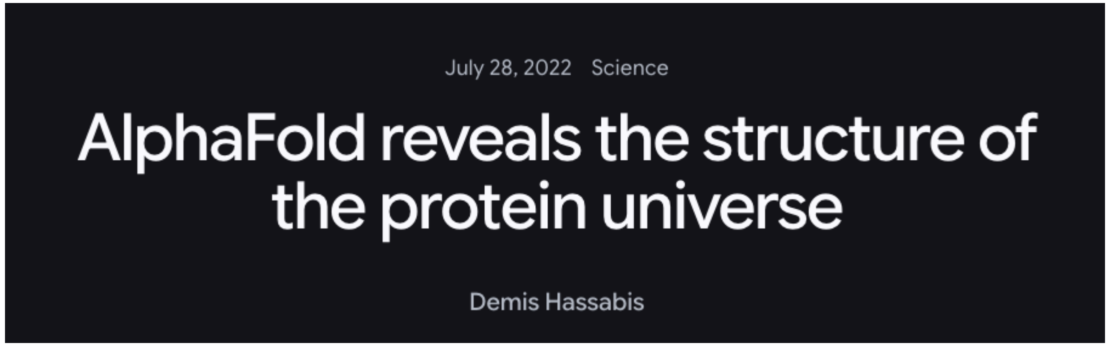
Additional reading **Key literature** **Links to Jupyter notebooks**

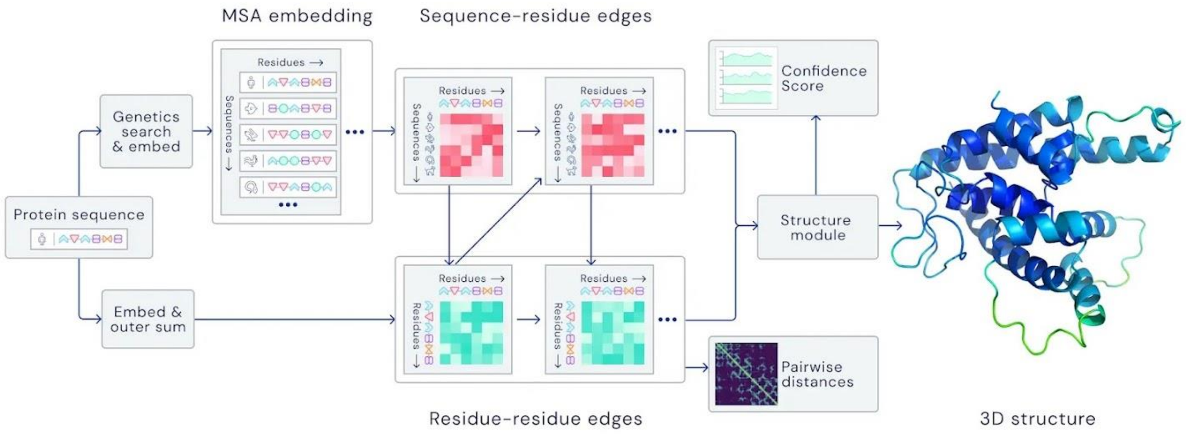
A brief history of machine learning in the chemical sciences (for artificial neural networks)

- ***Machine learning in chemistry is not new!***

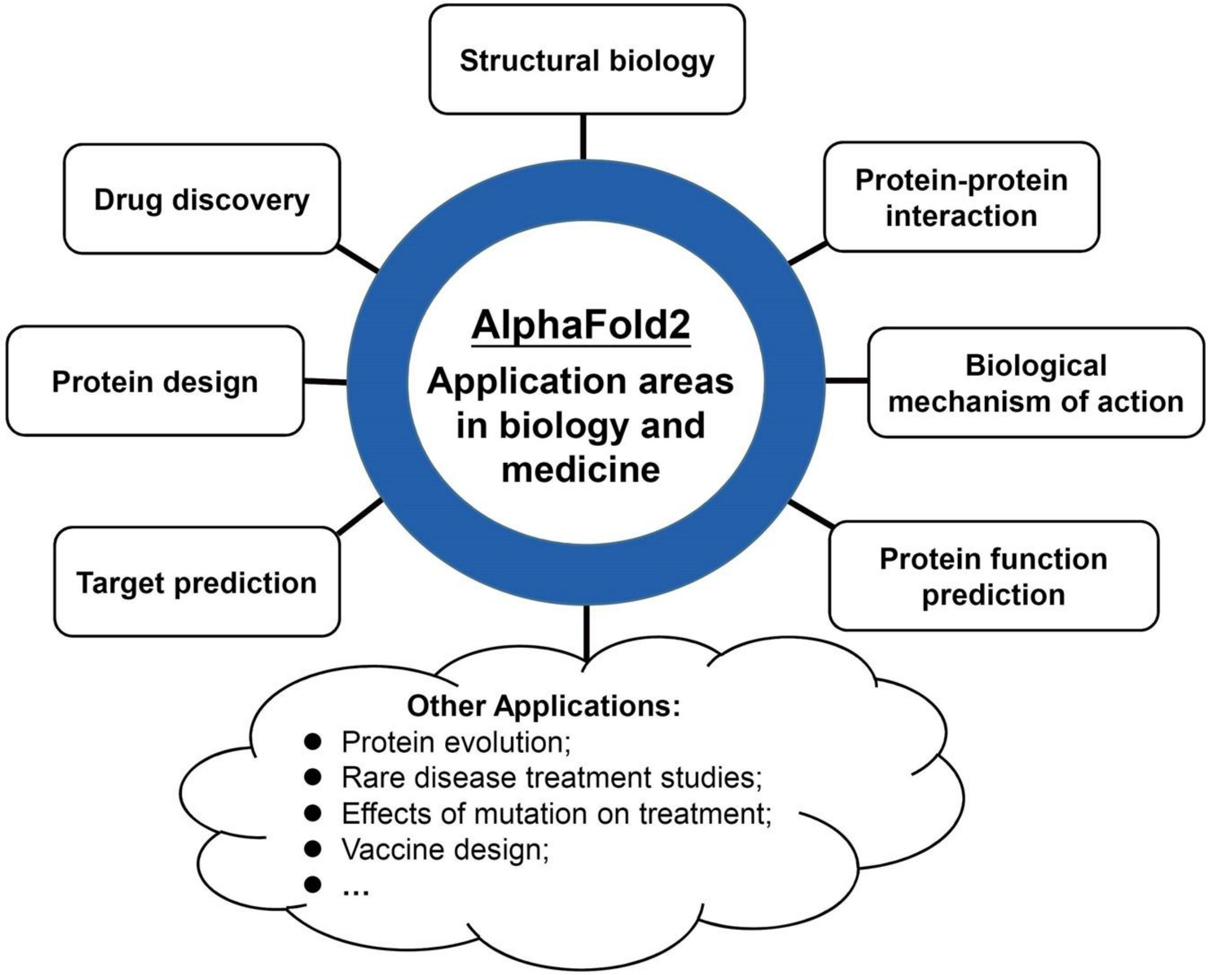


What has machine learning ever done for us?





- Alphafold protein structure prediction: [https://deepmind.google/science/alphafold/]
- Awarded **2024 Nobel Prize in Chemistry** ("*computational protein design*" – David Baker; "*protein structure prediction*" – Demis Hassabis and John Jumper)



What could machine learning do for me?

- What are the big outstanding challenges in your field where prior theoretical, computational or experimental methods have failed?
- What experiments or computations traditionally take a long time to complete but are still necessary?
- Which relationships between structure and experimental outcomes are complicated in a way that normally requires years of expert knowledge to infer?

Machine learning could help answer these questions….

Importance of high-quality datasets for machine learning

How "clean" or "polished" are the data?
Are there missing values?
Inconsistent naming/identifiers?
Duplicate entries (e.g., SMILES should be canonical)?
Is there good documentation?
Consistent file formats? (CSV, JSON, SMILES, etc.)
Experimental or computational error known?

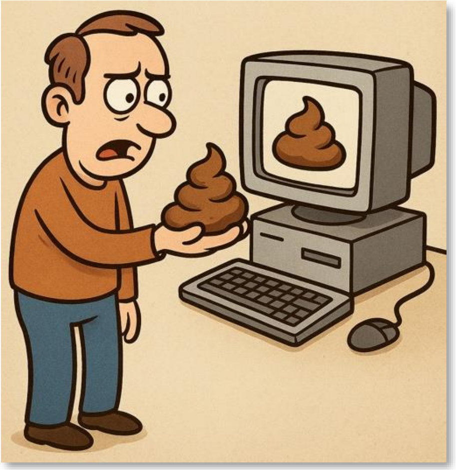
Preprocessing pipeline 1:
- Missing data handling
- Initial feature extraction and selection

Raw data collection → **Dataset** → **Training dataset** → **Processed training dataset** → **Machine learning algorithm** → **Final predictive model**

New dataset

Where does the data come from? Is it reliable?
From an experiment or computation?
Generated "in-house" or from elsewhere?

Preprocessing pipeline 2:
- Feature scaling
- Dimensionality reduction:
    - Feature selection
    - Feature extraction

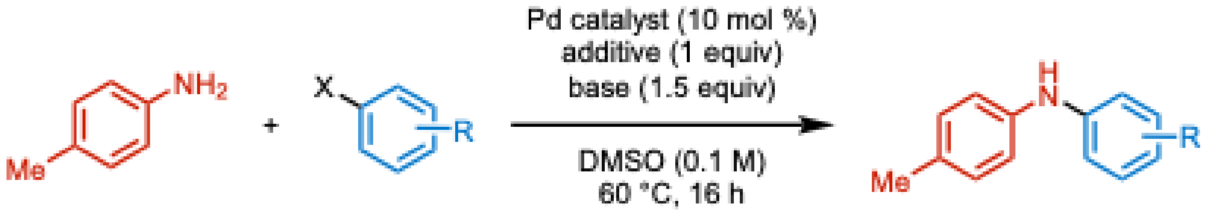


Doesn't machine learning require a lot of data?

- In general, **yes**, **but it depends!**..... And it's not as much as you might think…..

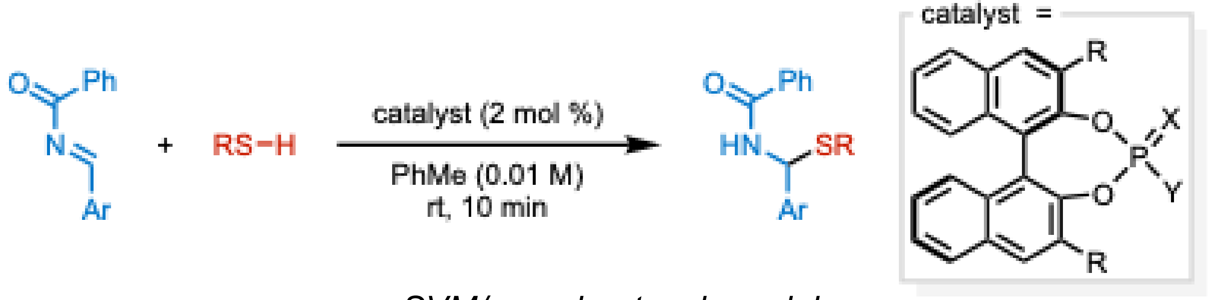
- Machine learning in chemistry often operates in a "**small data regime**" compared with other ML/AI fields but reliable models still require **100s–1000s of datapoints for most tasks**.

Doyle and co-workers, *Science* **2018**, *360*, 186



Random forest regression model
4608 reactions. Measured output = % yield

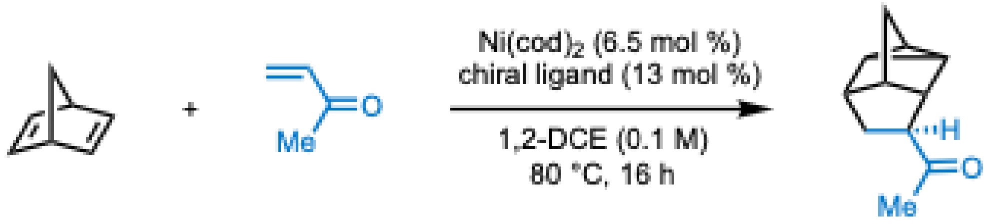
Denmark and co-workers, *Science* **2019**, *363*, 247



SVM/neural network model
2150 reactions. Measured output = enantioselectivity

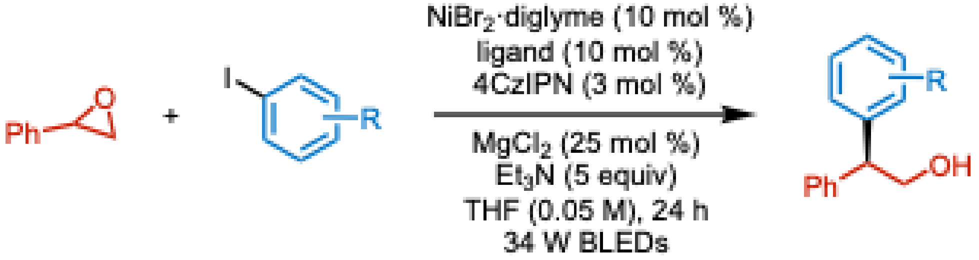
- Smaller datasets (<100) can be modeled with appropriate feature engineering, hyperparameter tuning and robust model performance testing.

Sigman, Reisman and co-workers, *J. Am. Chem. Soc.* **2025**, *147*, 31175



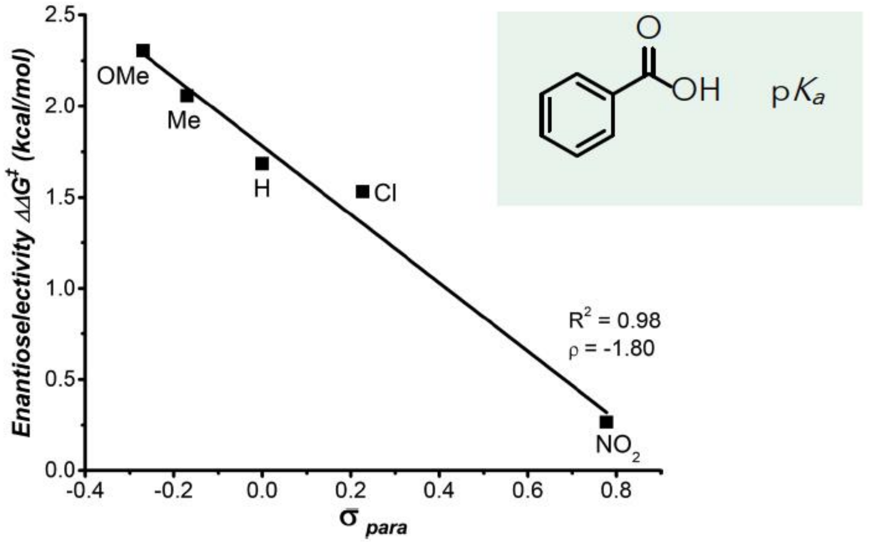
Linear regression model
12 reactions. Measured output = enantioselecivity

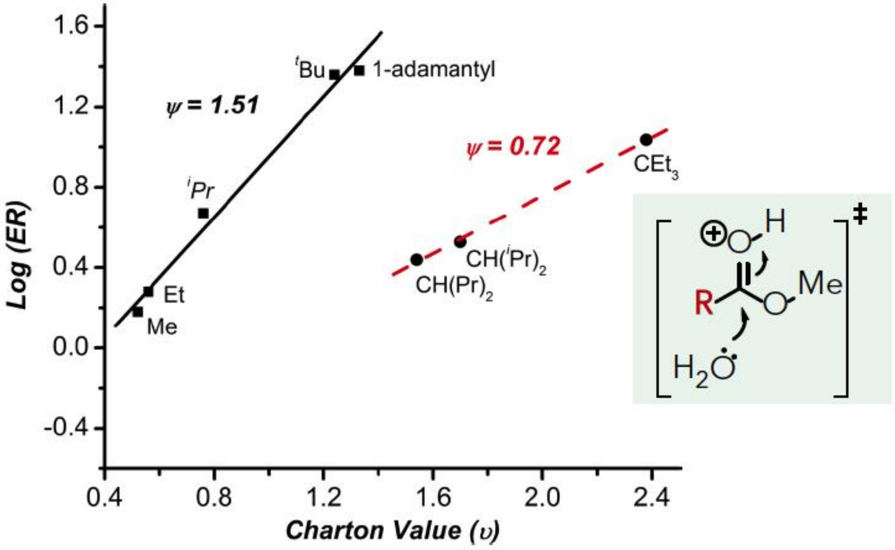
Sigman, Alegre and coworkers, *Chem. Sci.* **2025**, *16*, 8555 (original data: Doyle and co-workers, *J. Am. Chem. Soc.* **2021**, *143*, 15873)



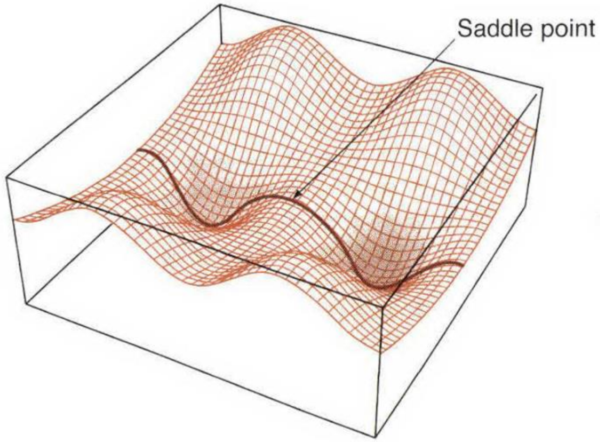
Neural network model
29 reactions. Measured output = enantioselectivity

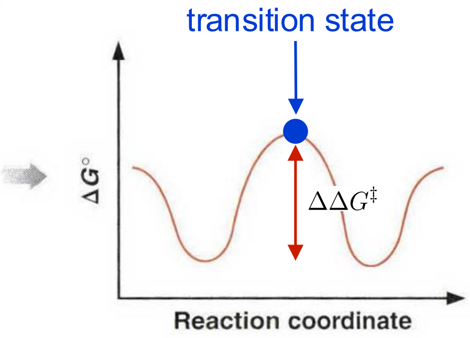
Reaction outputs used in chemistry modeling





- Quantitative structure activity relationships (QSAR) are informative about the mechanism of a reaction.
- Changes in electronics or sterics along a reaction coordinate can be probed.
- Linear free energy relationships describe the changes in activation free energy ($\Delta G^\ddagger$) or the reaction free energy change (ΔG) induced by a substituent.
- Substituent effect is described by a molecular feature (*e.g.*, electron-rich *vs* electron-poor or big *vs* small).



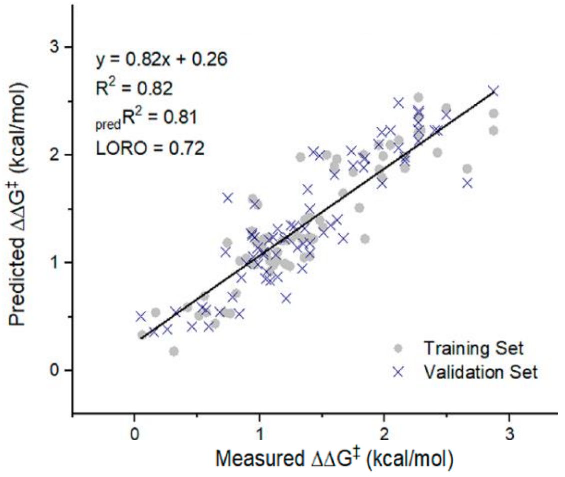


- $\Delta G^\ddagger$ can be obtained from rates (Eyring-Evans-Polyani equation) or from computation of a transition state. Also, er and dr can be converted to $\Delta\Delta G^\ddagger$.

$$k = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^\ddagger}{RT}\right)$$

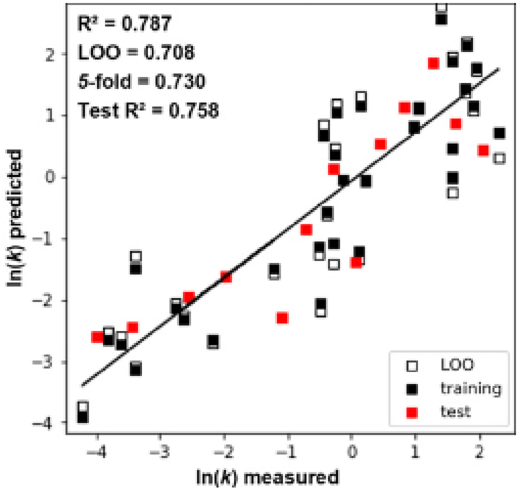
Reaction outputs used in chemistry modeling

- From transition state theory, we know that **$\Delta G^{\ddagger}$ or $\Delta\Delta G^{\ddagger}$ is a direct measure of reactivity**. Because regression models of this type of data are broadly **linear free relationships, they often give the most success in modeling**. They are also (mostly) **interpretable**!



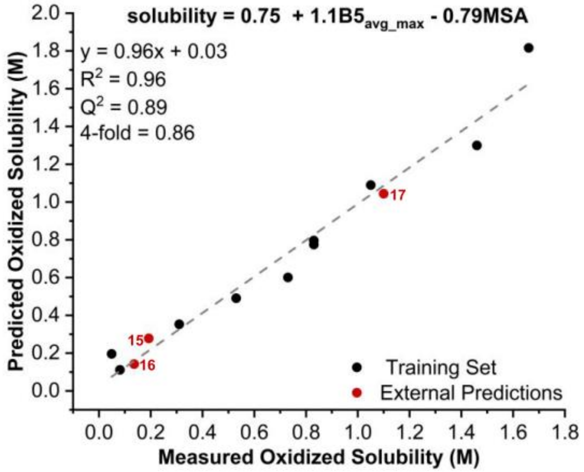
selectivity

J. Am. Chem. Soc. **2020**, *142*, 16382



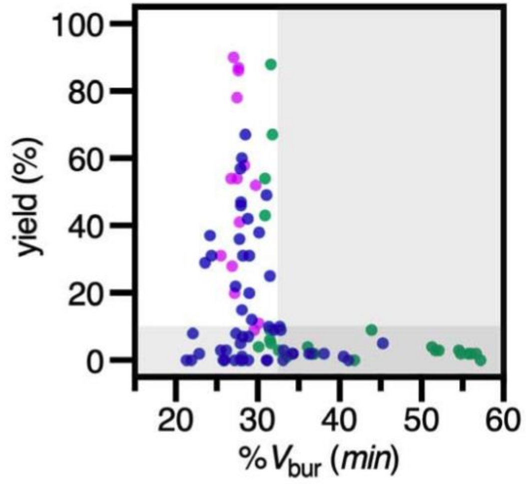
rates

Proc. Natl. Acad. Sci. U.S.A. **2022**, *119*, e2118451119



solubility

J. Am. Chem. Soc. **2021**, *143*, 13450



yield

Science **2021**, *374*, 301

- Reaction outputs which are not directly coupled to reactivity can cause issues. **Yield isn't necessarily a readout of reactivity** – **it can be made up of several factors**. As a result, modeling is usually **very challenging**!

Where will my experimental data come from?

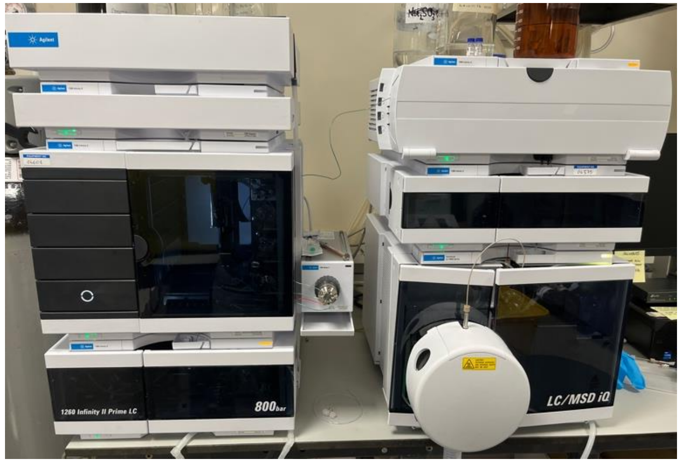
High-throughput experimentation facilities at Bath

Block heater / tumble stirrer for 96- and 48-well plates

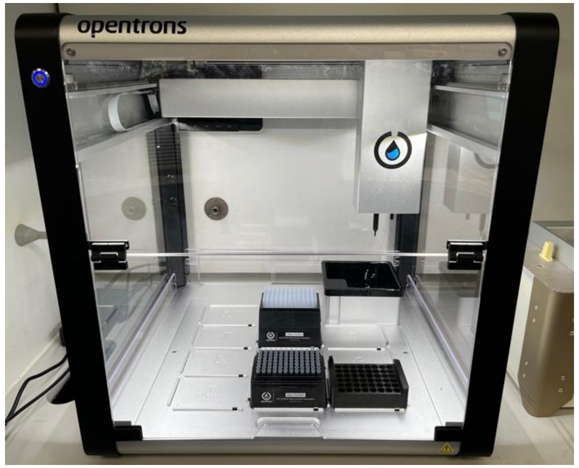
96-well LED arrays for photochemistry [420 nm & 365 nm]

Orbital heater/shaker for enzyme-catalyzed reactions

Custom plate design (CAD) and fabrication – 3D printing or CNC



Agilent HPLC-MS configured for analysis of 96-well plates + reaction monitoring capability



Opentrons OT-2 liquid handling robot

- Or you can get it from the literature…

Science
JACS
ACS Catalysis

+ other chemical reaction databases (USPTO, Pistachio, etc.)

- experimental errors
- experiment selection bias
- result reporting bias

Data biases in chemical reaction data!

Importance of high-quality datasets for machine learning

- Open access research data sharing is crucial for facilitating **scientific advancement**.

  - Data should be **FAIR**!

  Findable **A**ccessible **I**nteroperable **R**eusable

- **Findable**: should be able to find the data using a process that is easy for both humans *and* computers.

- **Accessible**: once data is found, the user needs to be able to readily access it (and know how).

- **Interoperable**: the data should be able to integrate with other data.

- **Reusable**: can be used for other purposes beyond the original one and the process of data production should be well described such that it can be replicated.

Why should I bother?

- Facilitates other downstream applications with the scientific community.

- Many journals require this now (especially the more ML-focused ones such as *Digital Discovery*).

- Grant applications usually require a data management plan.

- You can get credit for the data…. citations.

Where can I store my data?

- **GitHub/GitLab** – source code hosting and storage of CSV files.

- **Zenodo** – useful place for general file storage (CSV, computational files for a publication). Final dataset / code versions. DOI assignment - citable!

- **Hugging Face** – ML models and datasets. Ready to use ML models.

*Cambridge Crystallographic Data Centre - **Cystal structures***

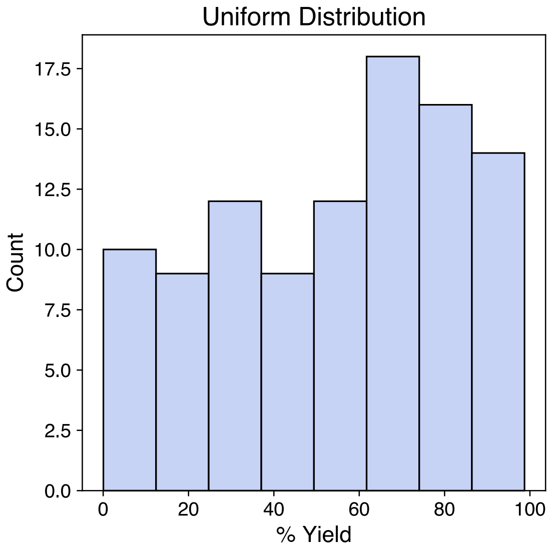
*ioChem-BD – **Computational data***

*The Materials Project – **Computed materials data***

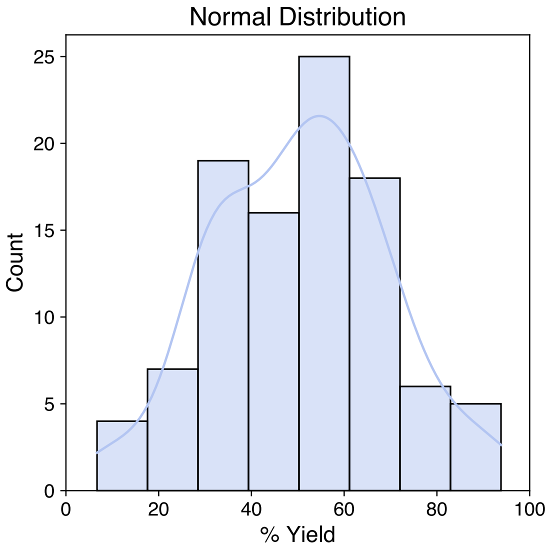
*Open Reaction Database – **Reaction data for machine learning***

Data distributions

- Before running any ML model, the **distribution of the data should be examined**. This is often represented using histograms which is a good way of **inspecting the quality of the data**. Also, might help to **decide what type of algorithm will be best** for modeling the data. Below are some common examples:

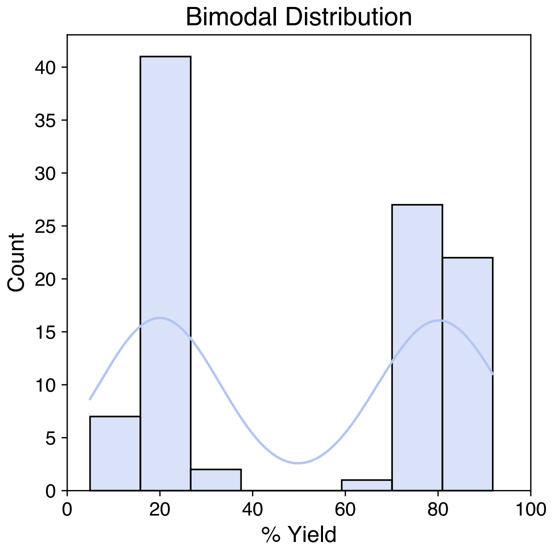


- All values have similar heights across the possible range.
- Each event in the interval is equally likely to occur.



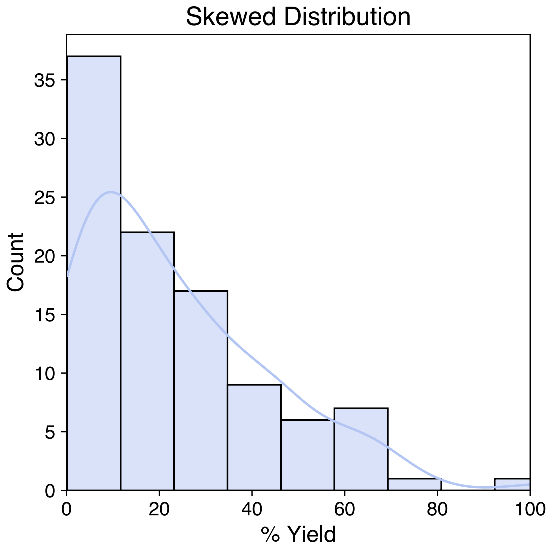
- Values are spread about the mean (50% yield in this case).
- Spread is controlled by the standard deviation – 68% of the data is in ±1 SD; 95% is within ± 2 SD.
- Linear regression assumes data is normally distributed.

Regression algorithms



- Data has two distinct peaks or modes – concentrated around two different values.
- Essentially have two distributions (or groups) of data.

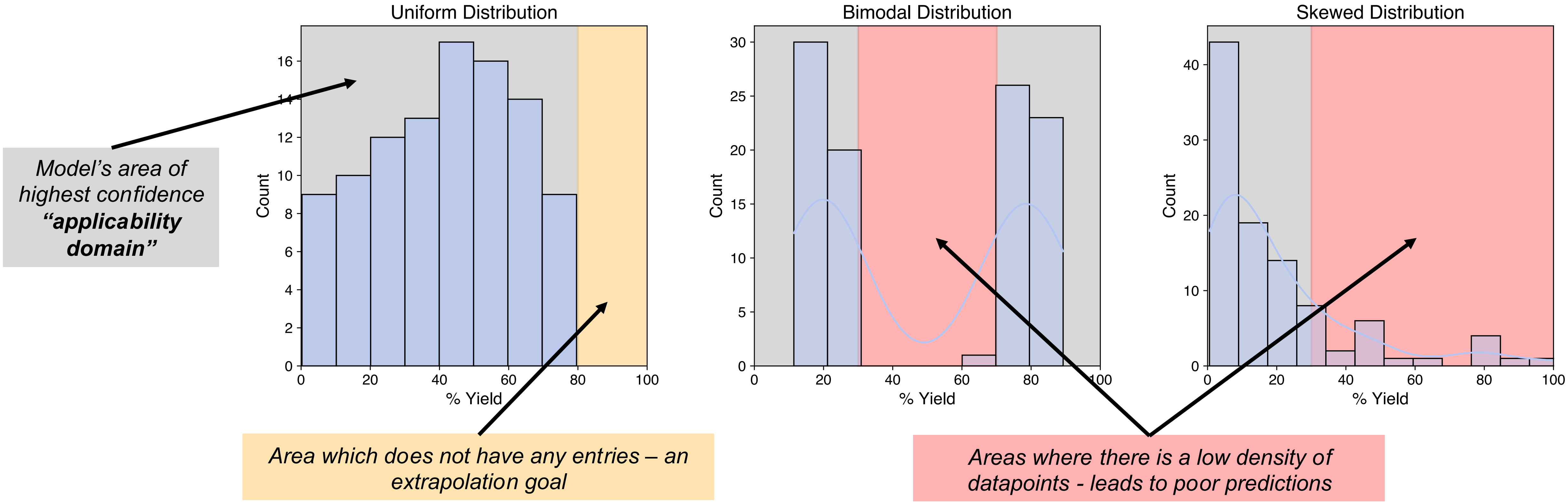
Classification algorithms



- Long tails on the left or right side of the data are characteristic of skewed distributions.
- Described as positive (long tail on right hand side) and negative (long tail on left hand side) skew.
- Regression algorithms can misestimate predictions.

Data distributions: Practical consequences for prediction

- How the data are distributed **can also affect the predictive power** of the ML model. Predictions can be **interpolations or extrapolations**.



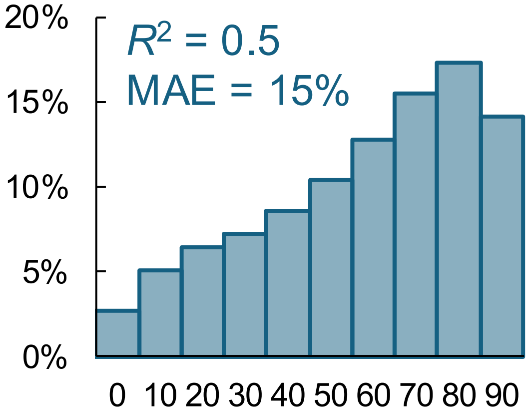
- An **applicability domain (AD)** of a quantitative structure activity relationship (QSAR) model is the **defined region where the model's predictions are reliable**. It represents **the chemical space covered by the training data** used to build the model.

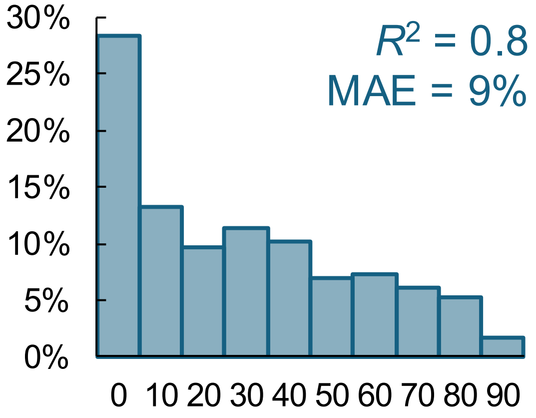
- A dataset with a **large range of values** in a uniform or normal distribution can be **reasonably extrapolated to give a higher** (or lower) **reaction output**.

- **Bimodel (or other multimodal) distributions can be dealt with using classification algorithms**. **Skewed distributions will almost always lead to poor predictions** (small AD) but **better dataset design** or use of **optimization algorithms** can lead to better predictions.

Matthew S. Sigman *et al. Sci. Adv.* **2025**, *11*, eadt3013

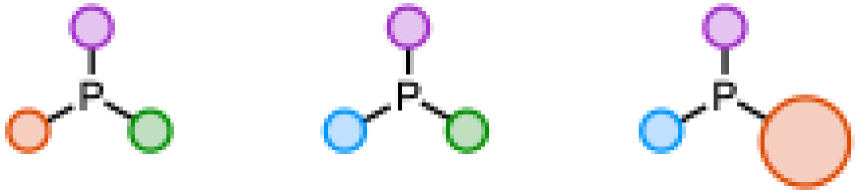
Data distributions: Practical consequences for prediction

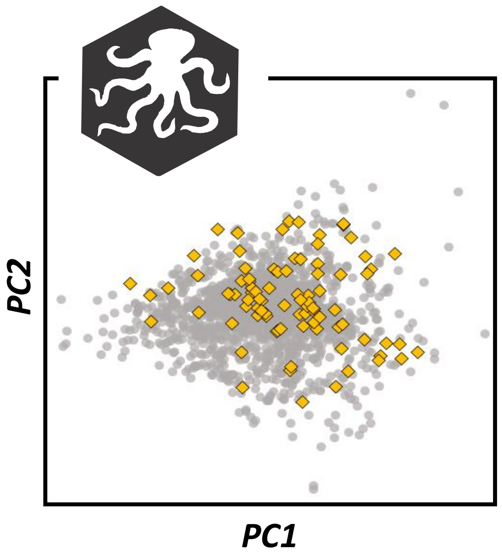




- Synthetic chemists are often tempted to report only those results which we **consider successful (high yielding reactions)** – the vast number of "**unsuccessful" experiments are not included**.
- When negative results are not included in a model, you **lose information about where reactions do not work** – *the learned reaction space becomes biased towards success*.
- **A model needs examples of both positive and negative outcomes to learn decision boundaries.**

- ~45,000 reactions were carried out using **triphenylphosphine**!
- **Cannot generate models describing how structural changes impact reactivity.**

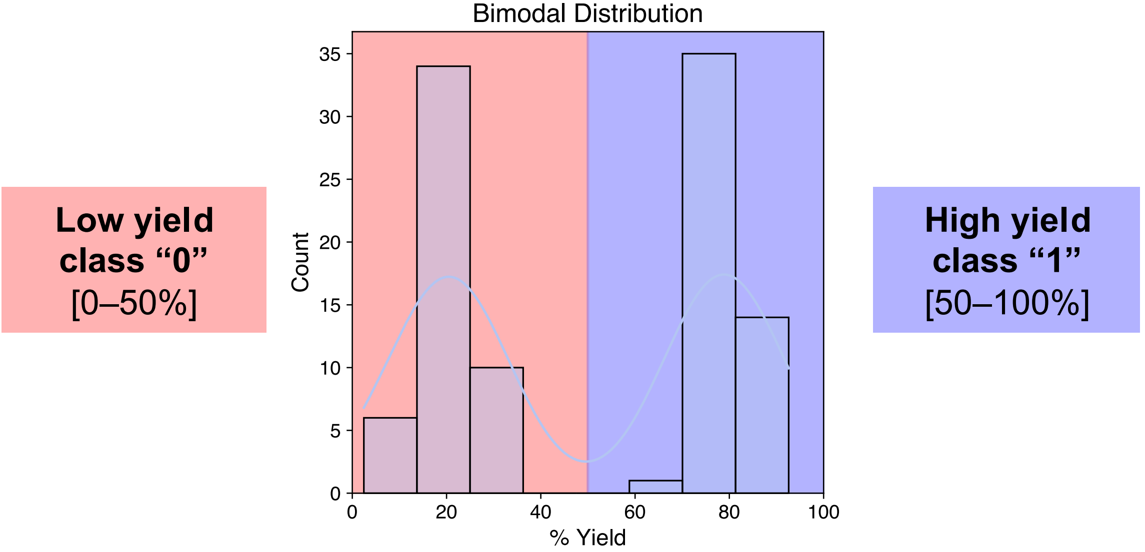




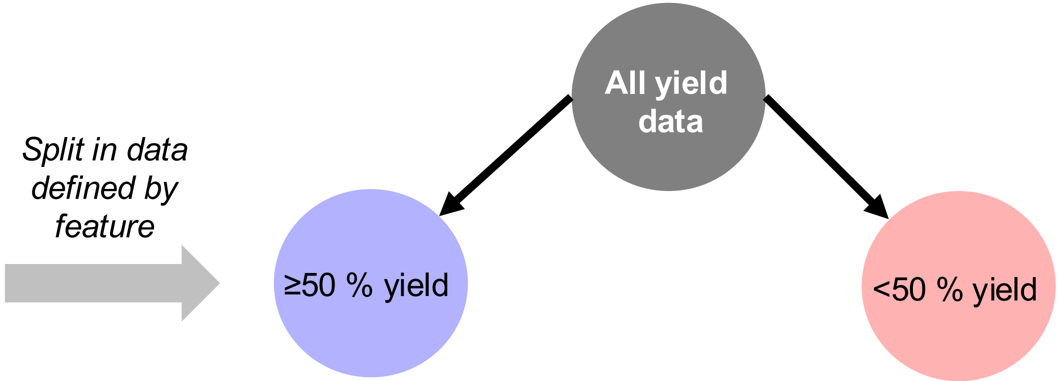
- A solution is to generate statistically diverse training sets to:
  - Enable trend identification
  - Boost hit likelihood
  - Increase knowledge gained
- Ligand libraries can help with diverse selection, see *kraken* (*J. Am. Chem. Soc.* **2022**, *144*, 1205; *ACS Catal.* **2022**, *12*, 7773) – **kraken.molssi.org**

Modeling categorical data: Classification algorithms

- If there is a split in the distribution of the data (bimodal distribution, for example), classification models usually work well.



- The simplest classification algorithm is a single node decision tree. The user defines where the data is cut and the algorithm finds a descriptor to best partition (or bin) the data.



- The algorithm iterates through all the features in the dataset and selects the best classification based on statistical metrics:
  - The **accuracy** is the fraction of correct predictions.
  - **Precision** is the ability of a classifier not to label as positive a sample that is negative.
  - **Recall** is the ability of the classifier to find all the positive samples.
  - The $\mathbf{F_1}$ **score** is the weighted harmonic mean of the precision and recall.

$$F_1 = 2\frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

TP = true positive; FP = false positive; FN = false negative

- The number of true positives, false negatives etc. can be easily visualized in a **confusion matrix**.

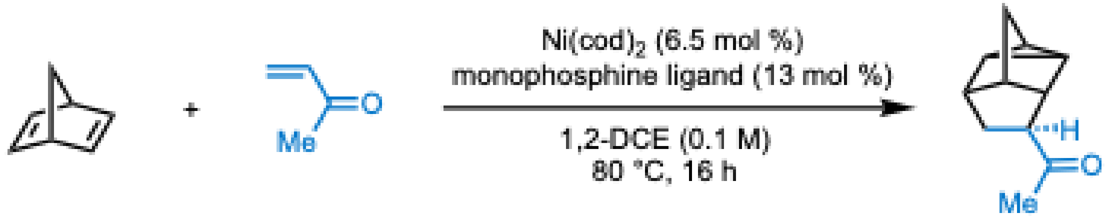
Predicted Values \ Actual Values	Positive (1)	Negative (0)
Positive (1)	TP	FP
Negative (0)	FN	TN

- For more complex problems you can imagine having a decision tree with more nodes – *i.e.*, more features.

- Equally, a classification algorithm may have more than two classes.

Example: generating a classification model

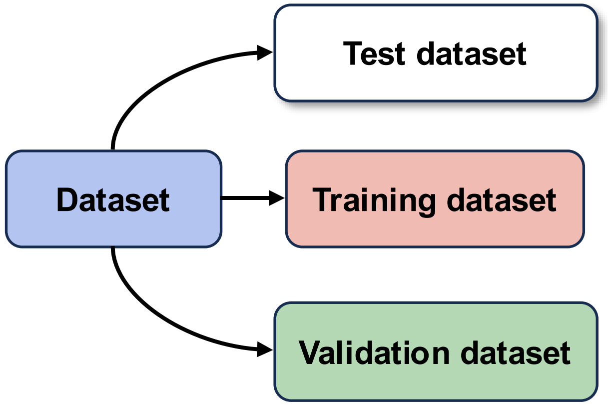
- Modeling the yield of a Ni-catalyzed homo-Diels–Alder cycloaddition (*J. Am. Chem. Soc.* **2025**, *147*, 31175)

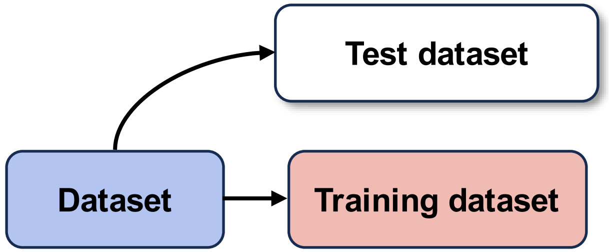


***Yield data** was collected for the cycloaddition between norbornadiene and methyl vinyl ketone with $Ni(cod)_2$ and **43 monophosphine ligands** (featurized using the kraken monophosphine library descriptors)*

Partitioning a dataset into training and test datasets

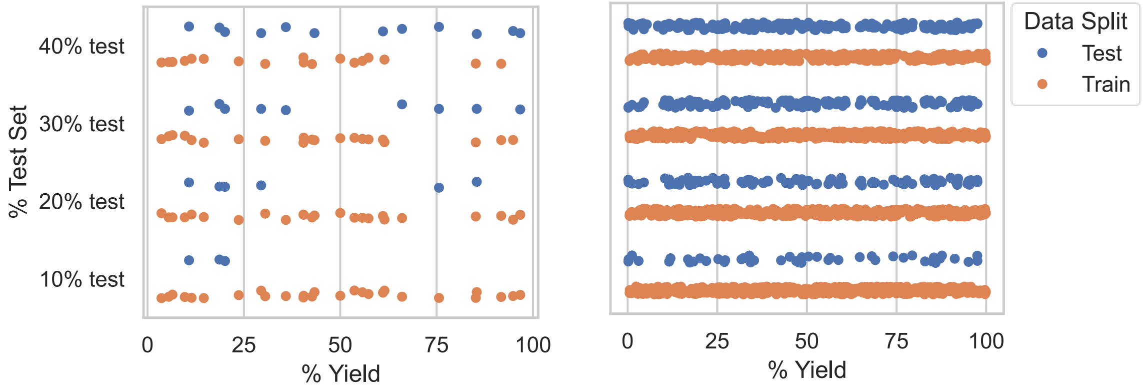
- Depending on the size of the dataset, we can opt for a two-way or three-way holdout method to divide our dataset.
- When there is a medium to large dataset size (for chemistry this is >100) a three-way holdout is used to both **validate** and **externally validate** the model. This is also common when tuning hyperparameters depending on algorithm choice (see later).
- A validation set is the data which is used to **evaluate the predictive power of the model** and is used alongside cross-validation methods.
- An external validation (test dataset) is used to **evaluate model performance on data totally unseen** (**unbiased**) to the model building process.

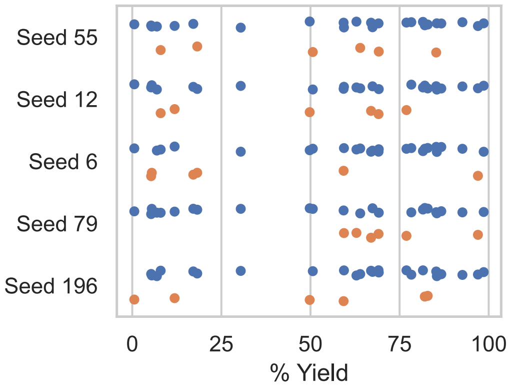




- The terminology used to describe these partitioned datasets can be slightly different (by subfield or author). We will generally define a two-way holdout having training+test datasets and a three-way holdout as training+validation+test.

- When dividing up datasets, remember that this involves **withholding valuable information** which could be used to train a model. Therefore, we don't want to put too much information into a test set.
- However, a **smaller test set size leads to a poor assessment of the model's predictive performance**. **Clearly there is a trade-off**.
- Below are two datasets (uniformly distributed) with 30 and 500 datapoints. A random split method with different test ratios was performed on each:

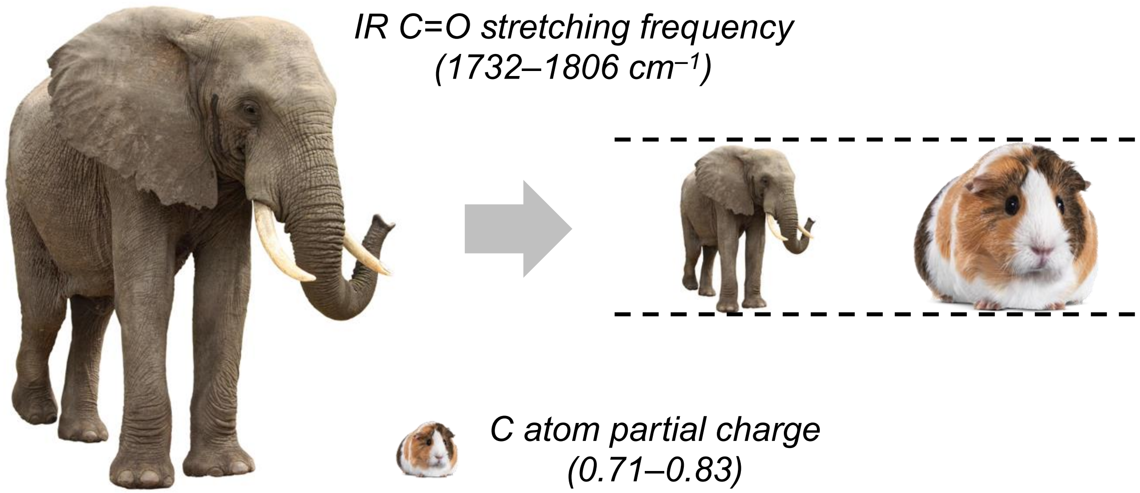




- Random split methods may not always be appropriate for smaller datasets (cannot guarantee good spread across distribution).
- Other methods such as y-equidistant, Kennard-Stone, k-means clustering are common alternatives.

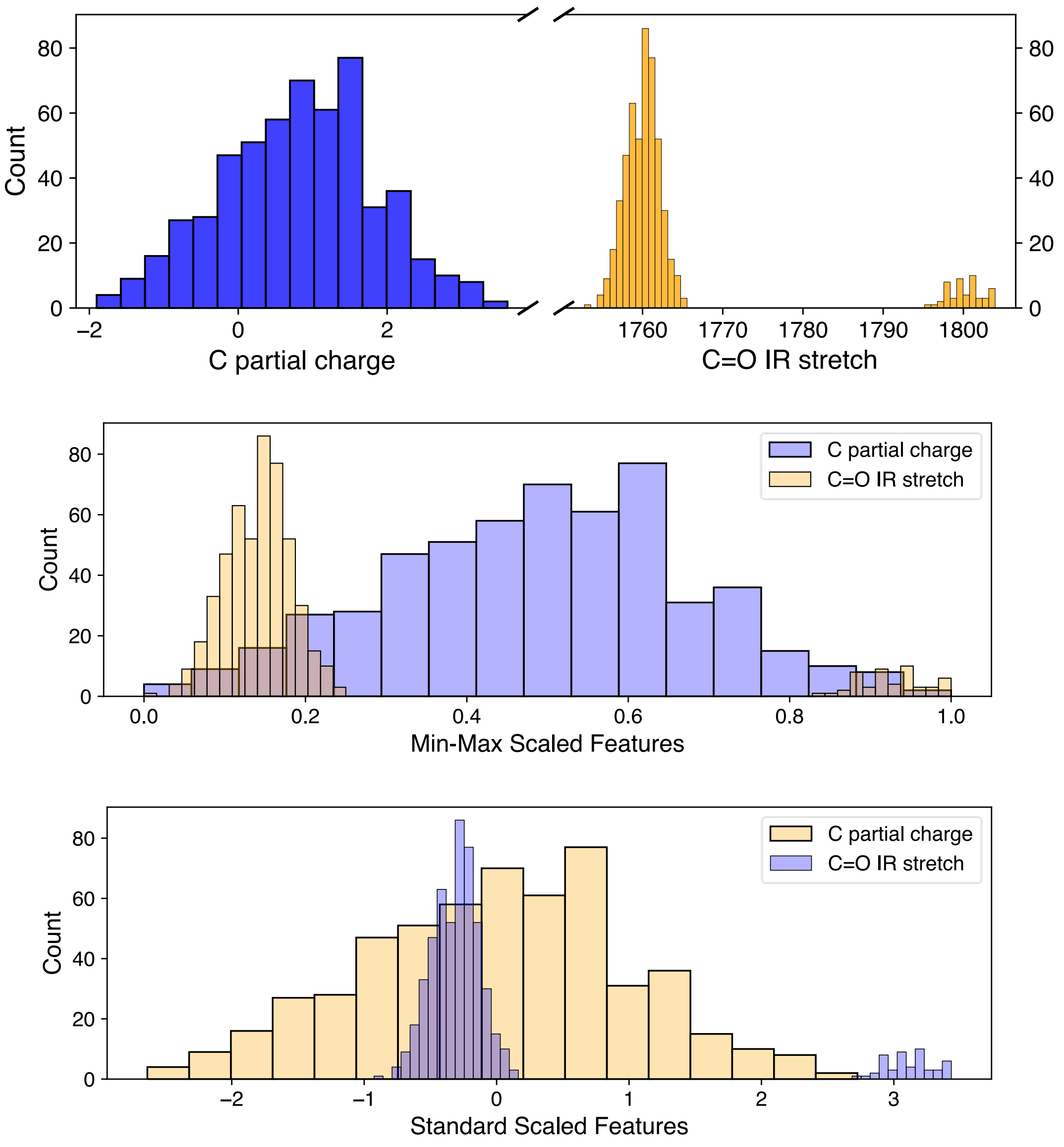
Bringing features onto the same scale

- Many algorithms (including linear regression) **require all the features to be on the same scale as each other**.
- Some algorithms are **scale-invariant**, where this is not a problem (*e.g.*, decision trees and random forests).



- Scaling prevents one feature from **outweighing the effects of another** in a model. In other words, allows us to compare the sign and magnitude of two features in a linear model equation.
- There are two ways in which features are commonly scales: **normalization** and **standardization**.
- Normalization refers to **rescaling of the features between 0 and 1** (min-max scaling).
- Standardization **centers features at mean 0 with a standard deviation of 1** – it does not change the shape of the distribution.

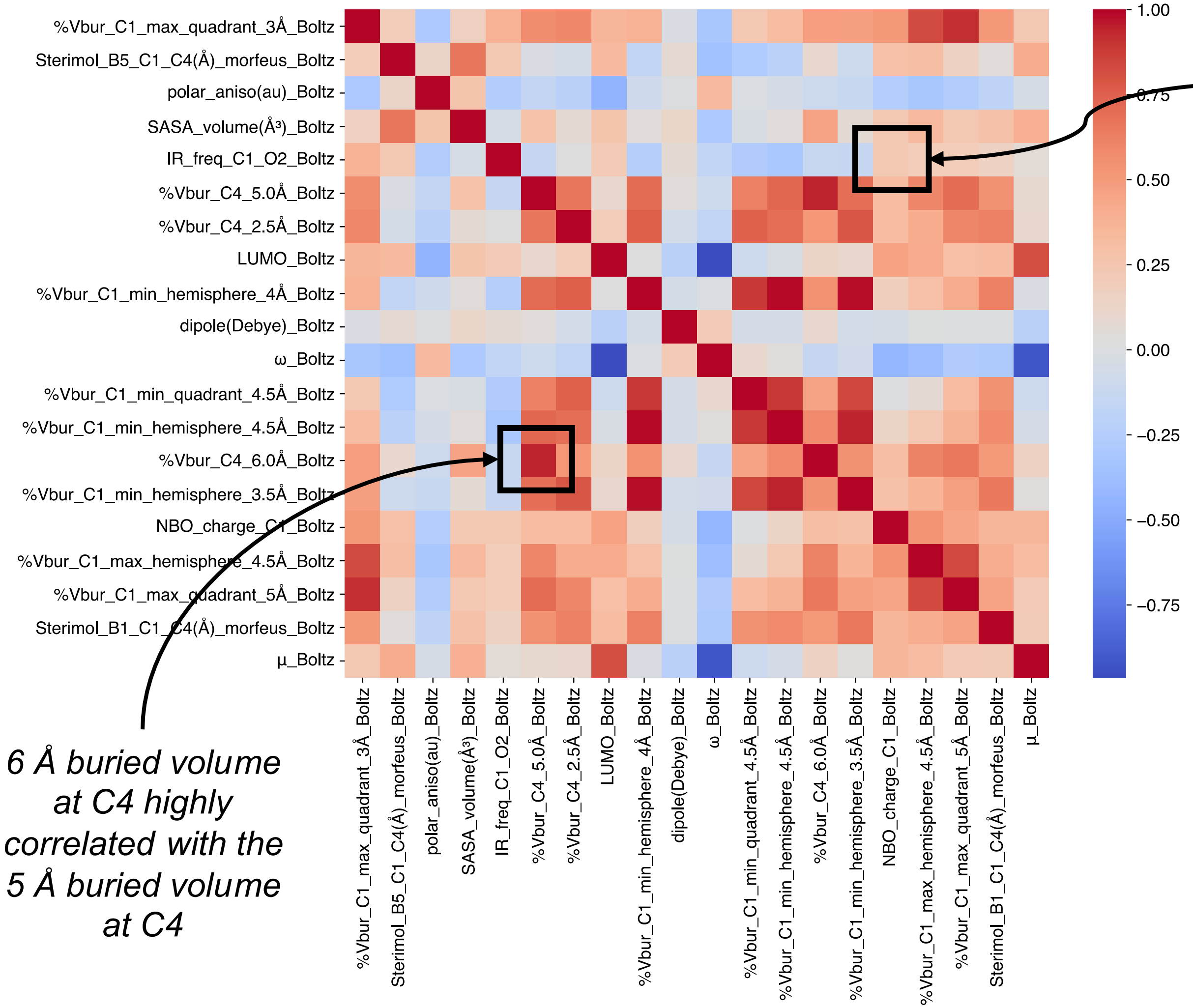
- Example with some randomly generated data:



Checking for correlations between features

- Looking for features which are **correlated** to one another is an **important filtering step** before using them in a machine learning algorithm. The correlations can be viewed in a **correlation matrix/heatmap**. The correlations are measured using the **Pearson correlation coefficient** (similar to R^2).

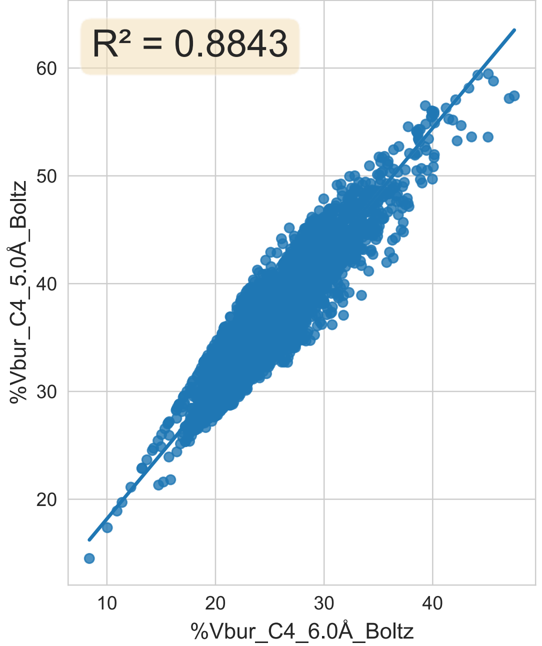
- This is done to prevent a model being trained on "duplicate information", *i.e.*, two features describing the same property of a molecule which can complicate the model building process and the interpretability.

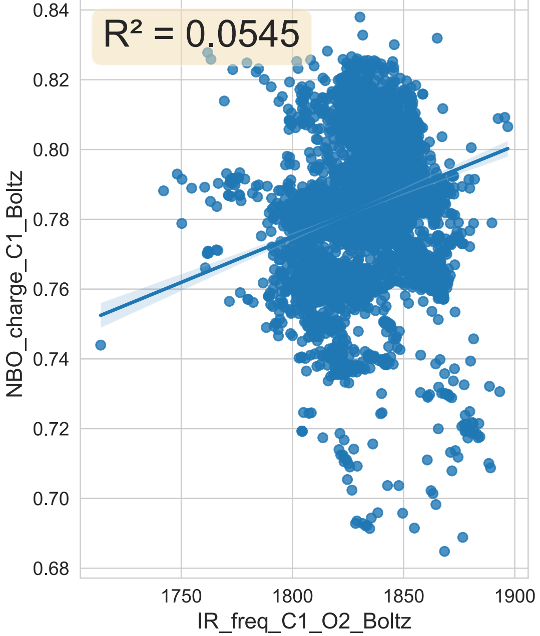


6 Å buried volume at C4 highly correlated with the 5 Å buried volume at C4

Much weaker but apparently positive correlation between C1=O2 IR stretch and the C1 NBO charge

- Looking at the direct correlation between two features can also be useful – this helps to visualize the overall trends.



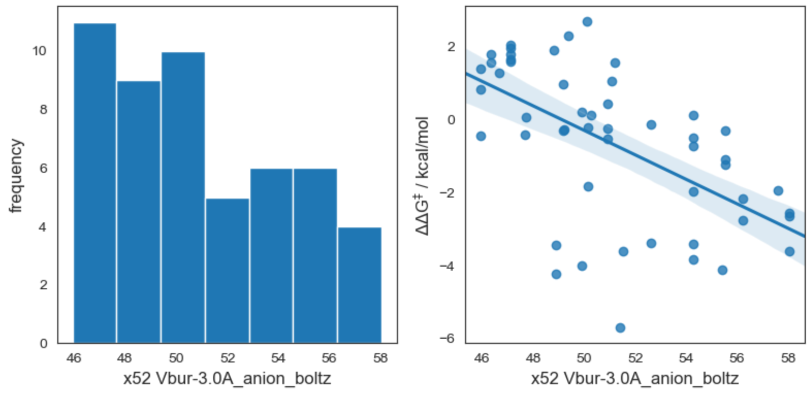


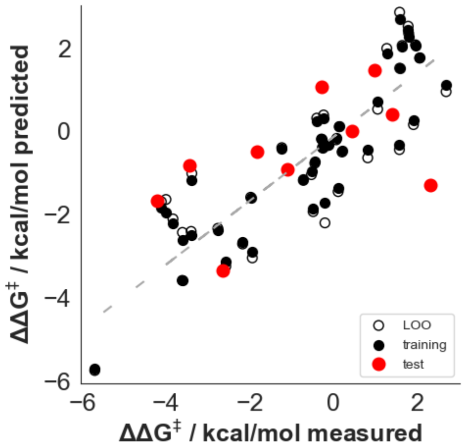
Linear regression modeling

- This is the simplest type of regression model which can often provide the most interpretability – they very often have similar characteristics of linear free energy relationships.

- The simplest type of linear regression model is a univariate linear regression. Which has the form:

$$y = mx + c$$





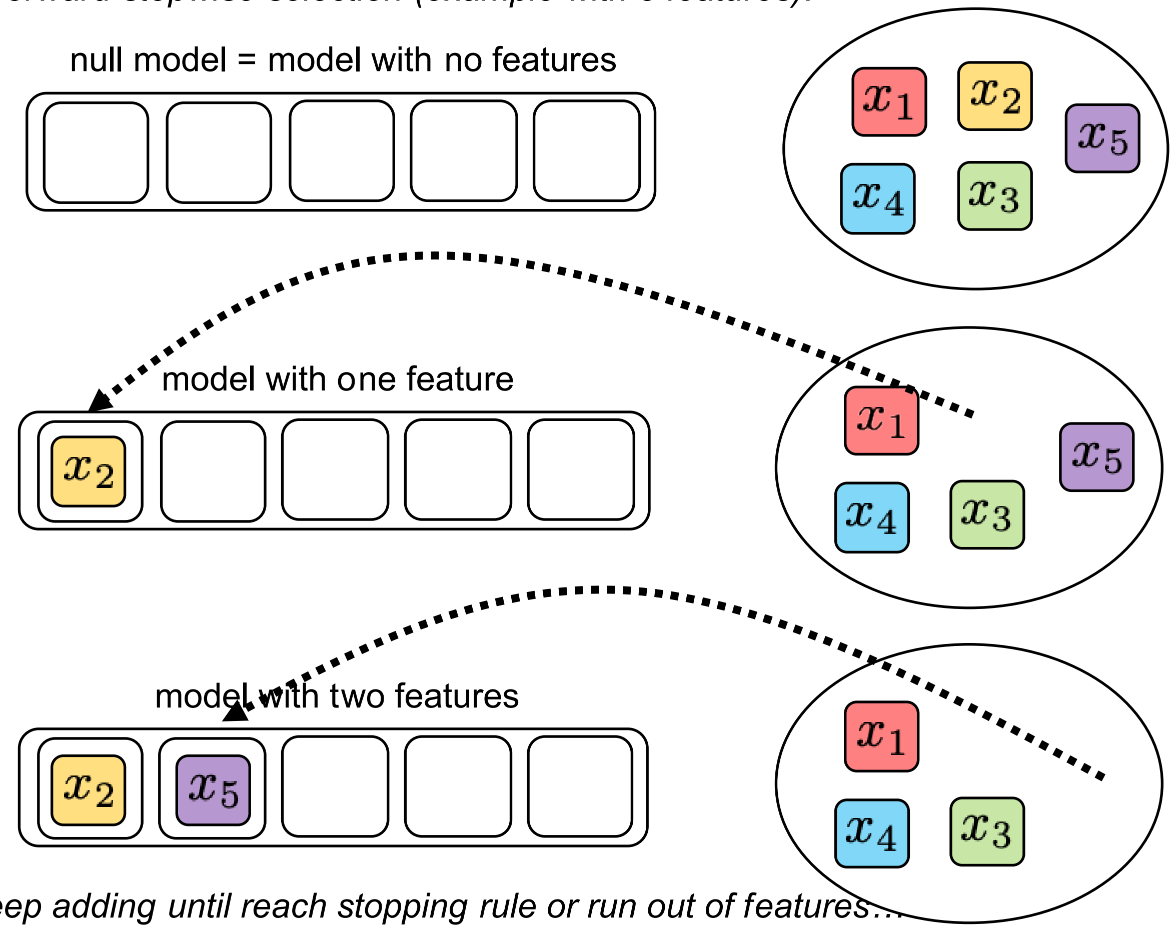
- A **multivariate linear regression (MLR)** (also called multiple linear regression) model is a more complex regression – which has **more than one feature** and has the form:

$$\underset{\text{target}}{y} = \underset{\text{intercept}}{\beta_0} + \underbrace{\beta_1 x_1 + \ldots + \beta_n x_n}_{\text{features and coefficients}}$$

- There are several ways in which features can be selected in an MLR model. One common process is **forward stepwise selection**. Others are **backward stepwise selection** and **brute force**.

- When there are many features to consider in a model building process, these feature selection methods help reduce the computational load.

Forward stepwise selection (example with 5 features):

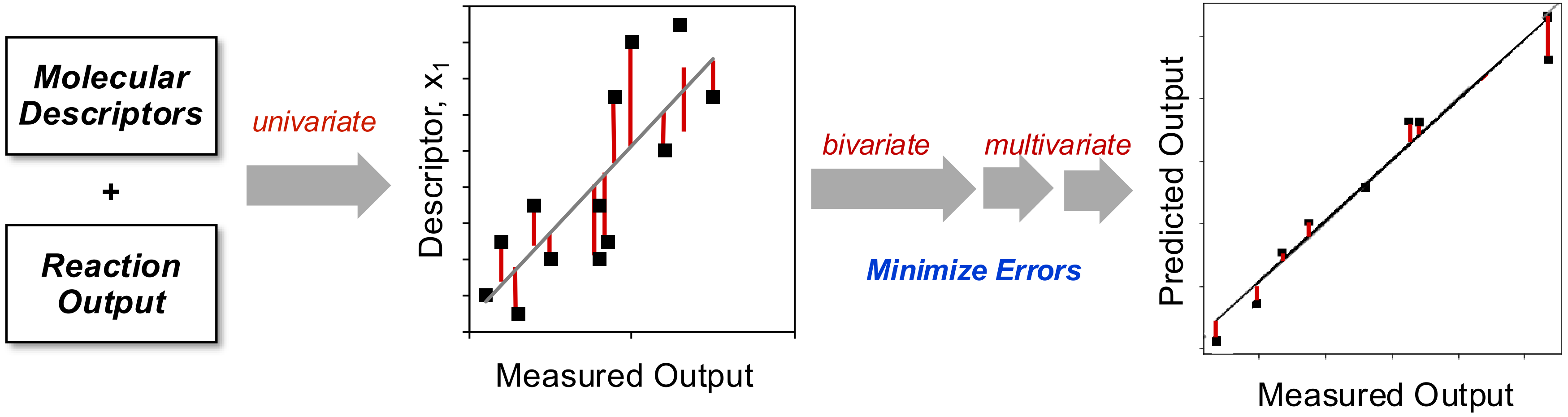


Keep adding until reach stopping rule or run out of features…

Linear regression modeling, continued

- Using a feature selection method such as forward stepwise helps to lower the computational cost of model building – but how do we know what features to add into the model?
- In each step of the algorithm a certain number of combinations of features are tested and those which give the lowest error get brought through to the next round.

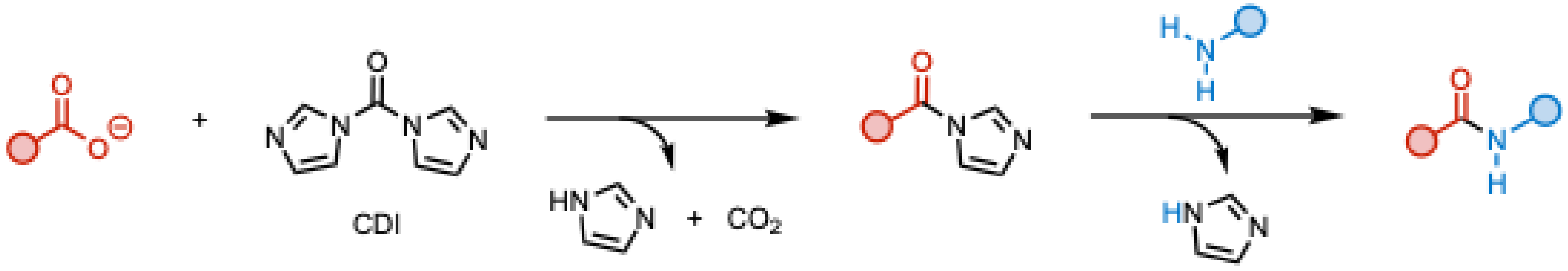
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots$$



- **How many features are warranted in an MLR model?** It is case dependent, but a good rule of thumb is ~10 training data points per feature in a model.
- What we put in (molecular descriptors and the reaction) determines what we get out! **Quality in = Quality out**.

Example: generating a linear regression model

- Modeling the rate of an amide coupling (*Proc. Natl. Acad. Sci. U.S.A.* **2022**, *119*, e2118451119)

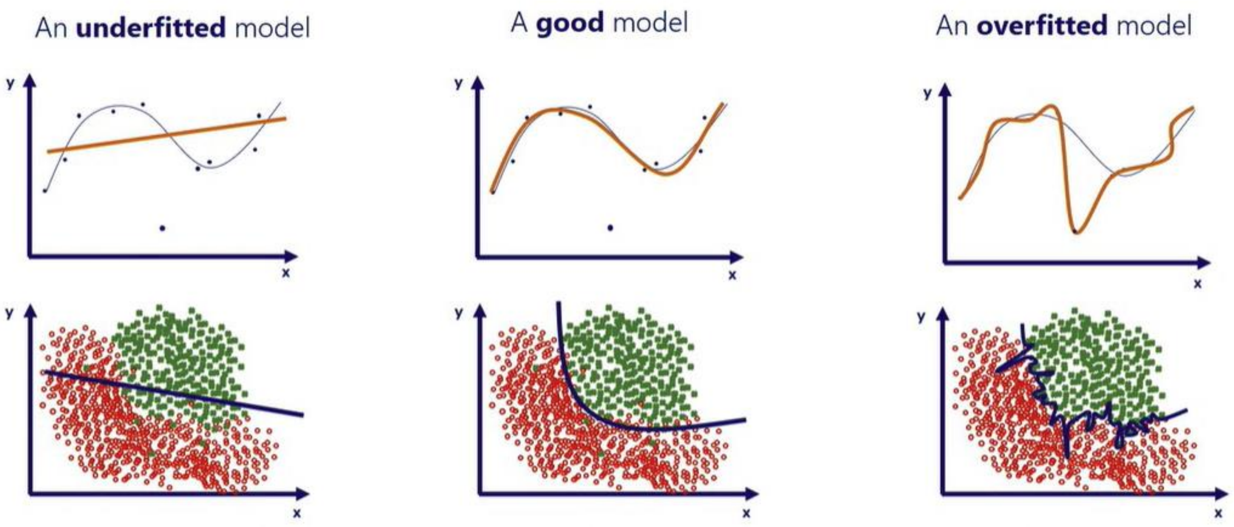


***Rate data** was collected for a series of amide coupling reactions. The individual acids (carboxylates) and amines were featurized to build a linear regression model containing descriptors of both components.*

How to select the best model

- The selection of the best model can be done **practically** or **statistically**. Practical model selection involves choosing a model which is more chemically interpretable.

- **Statistical selection is important** (more important than chemical selection, at least for an initial selection) to determine the **predictive ability**. Also determines the level of under- or overfitting to the data.



- In general, **underfit models have high training error and high test error**.

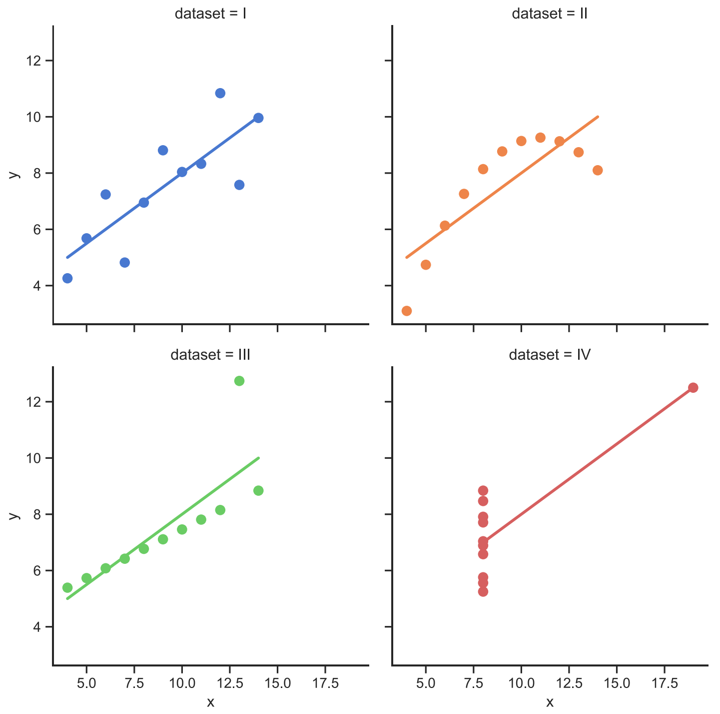
- **Overfit models have very low training error, but much higher test error**.

- **Well fit models have very similar training and testing error**.

- There are two main statistical metrics to determine the quality of a model – ***R*² (the coefficient of determination) and mean absolute error (MAE)**.

- R^2 represents the **proportion of variance** (of y) that has been explained by the independent variables in the model. It provides an indication of **goodness of fit** and therefore a measure of **how well unseen samples are likely to be predicted** by the model.

- **However, beware of R^2!** A high R^2 value may not always mean you have a good fit. See the below examples, all which give the same value of R^2 (this is Anscomb's quartet)!



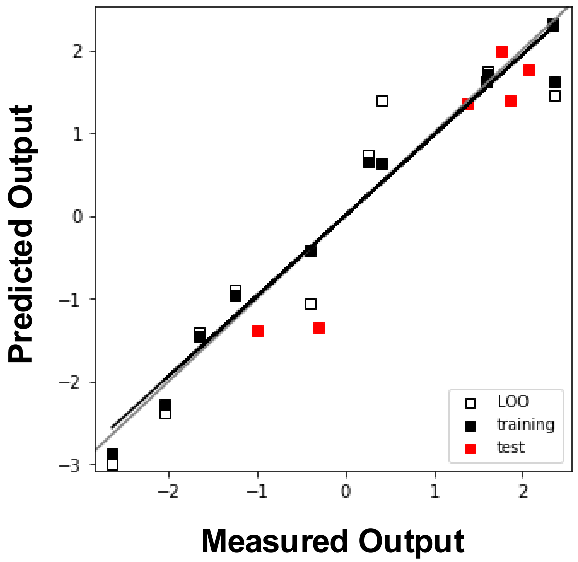
- **MAE is less susceptible to these issues**. And the units are in the same as the target.

- MAE is a risk metric corresponding to the error between paired observations expressing the same phenomenon. **In other words, the error between a true and predicted value**.

- Another example of a metrics is **root mean squared error** (**RMSE**).

How to select the best model, continued

- There are a number of ways to evaluate a model using statistical metrics such as R^2 or MAE.



- **Cross-validation** is used to determine how well the model will generalize to new data **within the training set**.

Training R^2 = 0.969
Training Q^2 (LOO) = 0.910
Training k-fold R^2 = 0.837
Test R^2 = 0.883

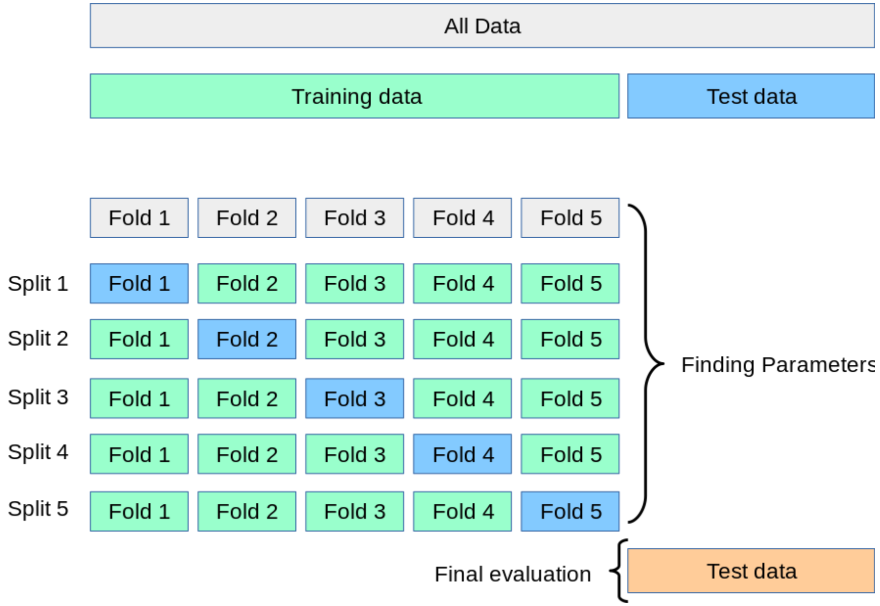
- These metrics are not restricted to R^2, they could also be MAE.

- **Leave-one-out (LOO) cross-validation** ensures **one datapoint is not biasing the model**:

Remove a data point → Refit model (only the coefficients change, not the features) → Predict removed data point (using the new equation) → Determine Q^2 (the fit of all the LOO predictions)

- ***K*-fold cross-validation** ensures a **group of data is not biasing the model**. LOO is the logical end of *k*-fold.

- Sometimes k-fold is not an appropriate cross-validation technique. If you have data which is **grouped by different catalysts or substrates**, you can employ a **leave-one-catalyst or leave-one-substrate out** cross-validation. This works in exactly the same as k-fold, except the folds are pre-defined.



- The **test prediction score** (R^2 or MAE) indicates **how well the test set** (*i.e.*, new, unseen data) **is predicted in the model**. This can also be used to measure the "**generality**" of the model.

- ***These validations and tests don't just apply to linear regression models; they are used in most types of machine learning model***.